

# Acoustic-Based Detection, Classification, and Localization of Impulsive Sounds

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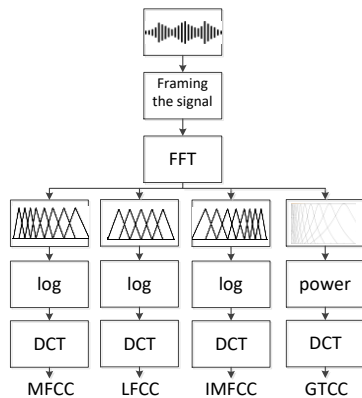
Due to the increasing frequency of shootings in public spaces and the rise in global gun ownership, there is a growing demand for enhanced autonomous security in communal areas such as campuses, hospitals, parks, and government facilities. While traditional surveillance technologies like cameras, drones, and security personnel provide some level of protection, they often fall short in high-risk situations involving active shooters. An automatic acoustic surveillance system presents a valuable addition, enabling faster and more precise detection of threats while reducing response times for security forces. Acoustic-based detection systems offer significant advantages by identifying gunfire, pinpointing its source, and tracking ongoing threats. These systems leverage the distinct acoustic signatures of firearms, using sound propagation and signal processing techniques to determine the location and caliber of a weapon.

The muzzle blast and the shock wave define a gunshot's acoustic pattern. The muzzle blast results from explosive gases expanding from the gun barrel, traveling at the speed of sound in the bullet's direction and lasting 1–3 ms, often resembling an "N" shape. If the bullet speed exceeds the speed of sound ( $v > c$ ), a secondary shock wave, or "N-wave," is produced due to compression and expansion shocks. Advanced signal processing is crucial to detect and classify gunshots accurately, as traditional time and frequency domain methods are unreliable. Spectral analysis often captures environmental noise rather than the gunshot itself, while time-domain patterns can be distorted by nonlinear dispersion and obstacles. Given these challenges, effective feature extraction must integrate time and frequency domain processing.

The proposed acoustic detection, localization, and classification system uses a distributed network of stand-alone sensors and a remote server. Each sensor continuously monitors its surroundings and transmits detected acoustic signals to a remote server for advanced processing. The system ensures broader coverage and adaptability to various environments by deploying multiple units. The system classifies acoustic events and triangulates their location using data from multiple sensors, enabling more precise and reliable 2D event localization. The system has been tested with various firearm calibers and ammunition.

For the identification of an event, an algorithm based on cepstral coefficients has been developed, see Fig. 1. Experiments revealed the effectiveness of this method as a feature extractor for further classification based on Neural Network (NN) and Support Vector Machine (SVM) classifier. The Mel-Frequency Cepstral Coefficients (MFCC) method, initially developed for other acoustic applications, has proven effective for gunshot recognition, with successful tests conducted in military shooting ranges. MFCCs are widely used for analyzing audio spectral characteristics, with modifications such as Linear-Frequency Cepstral Coefficients (LFCC) and Inverse Mel-Frequency Cepstral Coefficients (IMFCC) offering alternative filtering approaches. Other cepstral coefficients, like Linear Prediction Cepstral Coefficients (LPCC) and Perceptual Linear Prediction Coefficients (PLPC), serve various purposes, including noise elimination and speech recognition. Additionally, Greenwood Function Cepstral Coefficients (GFCC) and Gammatone-Frequency Cepstral Coefficients (GTCC), derived from human auditory response models, are used for classifying animal sounds and enhancing noise robustness in audio recognition, with GTCC offering greater resistance to noise compared to MFCC.

Experiments showed that MFCC, LFCC, and IMFCC yield similar results. At the same time, GTCC consistently performs better, as seen in Tab I. Increasing frame length improves classification accuracy but may also capture environmental reflections, potentially reducing focus on the gunshot itself. Results indicate that the NN classifier outperforms the SVM classifier due to its superior ability to model complex patterns. Overall, GTCC proves more effective than Mel-frequency cepstral coefficients, and NN achieves slightly better classification success, especially with longer frames.



| CLASSIFIER   |                   | SVM     | NN      | SVM     | NN      | SVM     | NN      |
|--------------|-------------------|---------|---------|---------|---------|---------|---------|
| FRAME LENGTH | FEATURE EXTRACTOR | ACC (%) | ACC (%) | RLC (%) | RLC (%) | MCC (-) | MCC (-) |
| 15 ms        | MFCC              | 83,90   | 85,59   | 80,65   | 67,74   | 0,62    | 0,62    |
|              | LFCC              | 83,90   | 83,90   | 80,65   | 64,52   | 0,62    | 0,57    |
|              | IMFCC             | 85,59   | 90,68   | 83,87   | 87,10   | 0,66    | 0,77    |
|              | GTCC              | 93,22   | 97,46   | 87,10   | 93,55   | 0,82    | 0,93    |
| 30 ms        | MFCC              | 84,75   | 87,29   | 67,74   | 87,10   | 0,60    | 0,70    |
|              | LFCC              | 87,29   | 83,90   | 70,97   | 70,97   | 0,66    | 0,59    |
|              | IMFCC             | 85,59   | 92,37   | 67,74   | 80,65   | 0,62    | 0,80    |
|              | GTCC              | 97,46   | 98,31   | 90,32   | 93,55   | 0,93    | 0,96    |
| 50 ms        | MFCC              | 84,75   | 93,22   | 74,19   | 90,32   | 0,61    | 0,83    |
|              | LFCC              | 88,98   | 86,44   | 77,42   | 77,42   | 0,71    | 0,66    |
|              | IMFCC             | 89,83   | 90,68   | 74,19   | 80,65   | 0,73    | 0,76    |
|              | GTCC              | 97,46   | 99,15   | 90,32   | 96,77   | 0,93    | 0,98    |

Fig. 1. The block diagram of the feature extraction algorithms.

Table I. Example of Classification results for a 9 mm class.